

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: MAT101R01

Course: ENGINEERING MATHEMATICS - I

QP No. : U019R-1

Duration: 3 hours

Max. Marks: 100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Find the angle of elevation x , of top of a tower from ground, when $\cos x = \frac{1681}{1682}$, assuming x is small.
2. If $\cos^{-1}\left(\frac{3i}{4}\right)$ represents a point in a two-dimensional space, then identify its abscissa and ordinate.
3. Find the algebraic and geometric multiplicities of the Eigen values of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
4. Test whether the static forces $F_1(1, 2, -1)$, $F_2(0, 3, 2)$, and $F_3(3, 0, 1)$ due to difference of weight and buoyancy acting on a ship are independent.
5. Find the rotated conic equation of $x^2 + 4xy + y^2 - 9 = 0$ after eliminating the xy -term.
6. Find the value of $\int_0^{\infty} \sqrt[4]{x} e^{-\sqrt{x}} dx$.

7. Find the value of $\int_0^{\pi/2} \sqrt{\tan x} \, dx$.

8. Determine $\iiint_R yz \, dx dy dz$ where $R: 0 \leq x \leq 1, 1 \leq y \leq 2, 2 \leq z \leq 3$.

9. Find any two roots of $8x^3 - 14x^2 + 7x - 1 = 0$, given the roots are in geometric progression.

10. When x is large, show that $\sqrt{x^2 + 16} - \sqrt{x^2 + 9}$ is inversely proportional to x .

PART - B

Answer all the questions

4 x 15 = 60 Marks

11. (a) If $\tan(\theta + i\phi) = \cos \alpha + i \sin \alpha$, prove that $\theta = \frac{n\pi}{2} + \frac{\pi}{4}$ and

$$\phi = \frac{1}{2} \log \tan \left(\frac{\pi}{4} + \frac{\alpha}{2} \right) \quad (8)$$

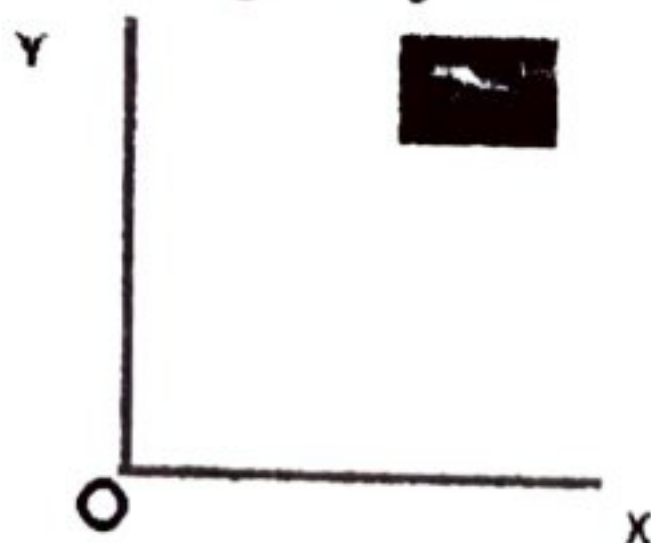
(b) Obtain the expansion of $\frac{\sin 7\theta}{\sin \theta}$ in a series of powers of $\cos 2\theta$

(7)

(OR)

12. (a) A shark in an Ocean is located at a point given by $i^{\log(1+i)}$ with respect to XOY coordinate system. Calculate the perpendicular distance from the Shark to imaginary axis

(8)



(b) Expand $\sin^3 \theta \cos^5 \theta$ in a series of Sines of multiples of θ

13. (a) A population of mice has the characteristics below:

A total of 75% of the population survives the first year. Of that 75%, 25% survives the second year. The maximum life span is 3 years. The average number of offspring for each member of the population is 2 the first year, 4 the second year, and 2 the third year. The population now consists of 160 members in each of the three age classes. How many members will there be in each age class in next two years? (8)

(b) Determine the values of a and b for which the system of equations: $x + y + z = 6$; $x + 2y + 5z = 10$; and $3y + 2x + az = b$ possess (i) a unique solution (ii) no solution (7)

(OR)

14. (a) Decompose the matrix $A = \begin{pmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{pmatrix}$ into $U \Sigma V^T$ by using singular value decomposition method (10)

(b) Balance the following chemical equation



15. (a) Solve the equation $8x^5 - 22x^4 - 55x^3 + 55x^2 + 22x - 8 = 0$ (8)

(b) Determine the sum of the infinite series

$$\frac{15}{16} + \frac{15.21}{16.24} + \frac{15.21.27}{16.24.32} + \dots \quad (7)$$

(OR)

16. (a) Apply Routh array method to identify the stability of the system represented by the following characteristic equation

$s^5 - 2s^4 - 10s^3 + 8s^2 + 33s + 18 = 0$. Also locate the roots of the equation (8)

(b) If a, b, c are the roots of the equation $x^3 + 2x^2 + 3x + 3 = 0$ then

calculate $\sum \frac{a^2}{(a+1)^2}$ (7)

17. (a) Evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dx dy$ by changing the order of integration.

Sketch the region of integration (8)

(b) Show that $\int_0^\infty x^n e^{-a^2 x^2} dx = \frac{1}{2a^{n+1}} \Gamma\left(\frac{n+1}{2}\right)$ and hence evaluate

$\int_0^\infty e^{-4x^2} dx$ (7)

(OR)

18. (a) Prove the relation $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence find $\Gamma\left(\frac{1}{2}\right)$ (8)

(b) Evaluate $\iiint_V dx dy dz$ where V is the finite region of tetrahedron formed by the planes $x = 0, y = 0, z = 0$, and $2x + 3y + 4z = 12$ (7)

PART - C

Answer the following

1 x 20 = 20 Marks

19. (a) Reduce the quadratic form $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ into a canonical form by orthogonal transformation and hence find its nature, rank, index, and signature (10)

(b) Determine the area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ using gamma function. (10)
